

DATA / MACHINE LEARNING / REAL ESTATE / SWISS OPEN DATA

PREDICTING APARTMENT RENTAL PRICES IN SWITZERLAND

A supervised machine-learning approach for the Swiss cold-rent market

We built a reproducible pipeline that scrapes Swiss rental listings, enriches them with official building (GWR) and swisstopo location data, and predicts the monthly cold rent of a single apartment. From 9'994 scraped listings, 4'536 fully enriched apartments were used for modelling. GradientBoosting with the ALL+geo feature set is the final model: not the lowest error, but the best balance of accuracy and generalisation.

01 THE PROBLEM

Can a single apartment's rent be judged fairly?

2-room flat · City of Lucerne

WHAT SHOULD IT COST?
Asking rents vary widely for similar flats



Location dependency

The same flat costs far more in the city of Lucerne than in rural regions.



Many interacting features

Area, rooms, building age and local market interact non-linearly.



No apartment-level benchmark

Official market indicators do not price one specific flat.

02 FROM DATA TO MODEL

From 25'002 listing URLs to 4'536 enriched apartments.

25'002 URLs identified
rentumo.ch, one scrape on 13 Apr 2026

9'994 listings scraped
detail pages successfully extracted

9'582 with core fields
address · area · rooms · cold rent

4'536 fully enriched
GWR building + swisstopo location data

ENRICHMENT STACK

rentumo.ch GeoAdmin GWR / BFS swisstopo

MODEL INPUT · 11 PREDICTORS + GEO CLUSTER

Apartment: area, rooms · Building: year built, dwellings, land area · Location: coordinates, elevation, population, public-transport score, solar class

03 RESULTS

Every tree model clearly beats the linear benchmark.

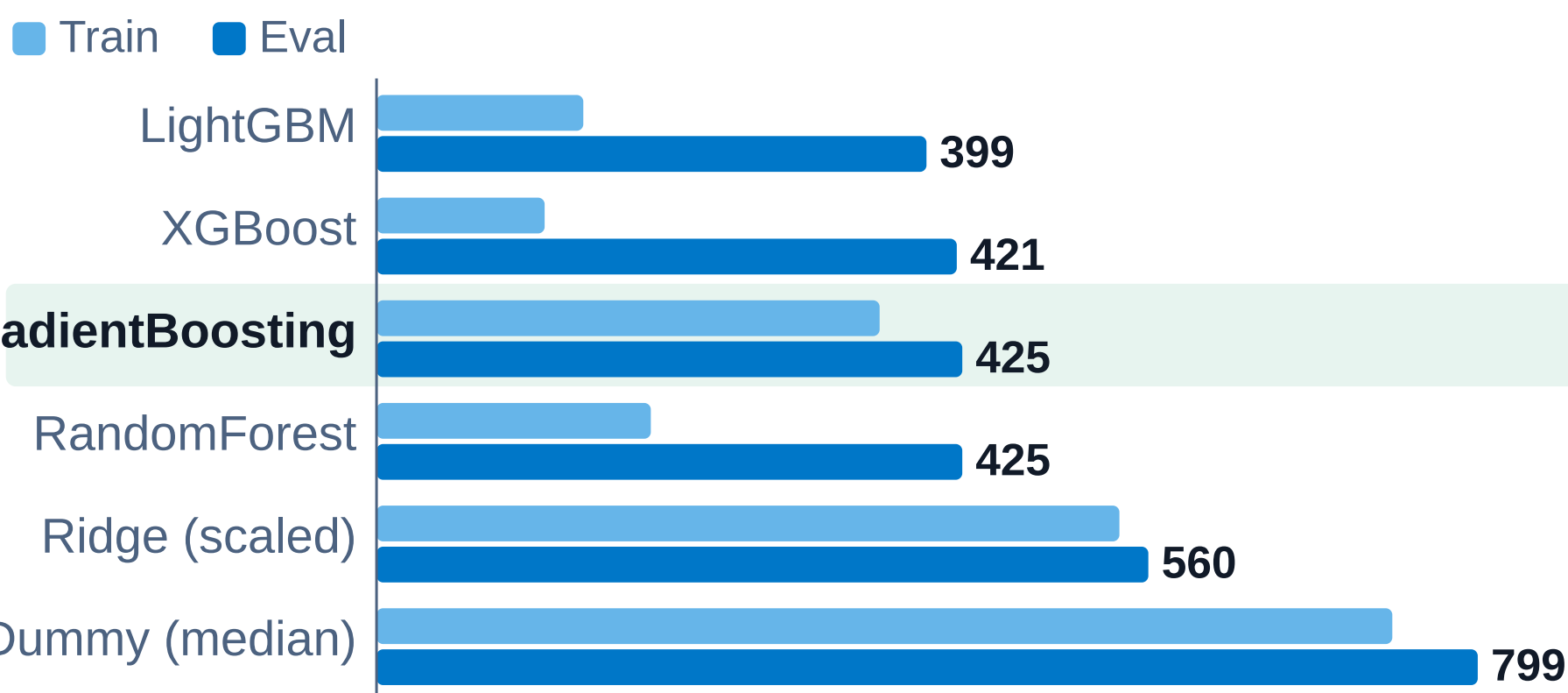
FINAL MODEL
GRADIENTBOOSTING · ALL+GEO

EVAL RMSE
CHF 425

EVAL R²
0.710

TRAIN-EVAL GAP
CHF 60

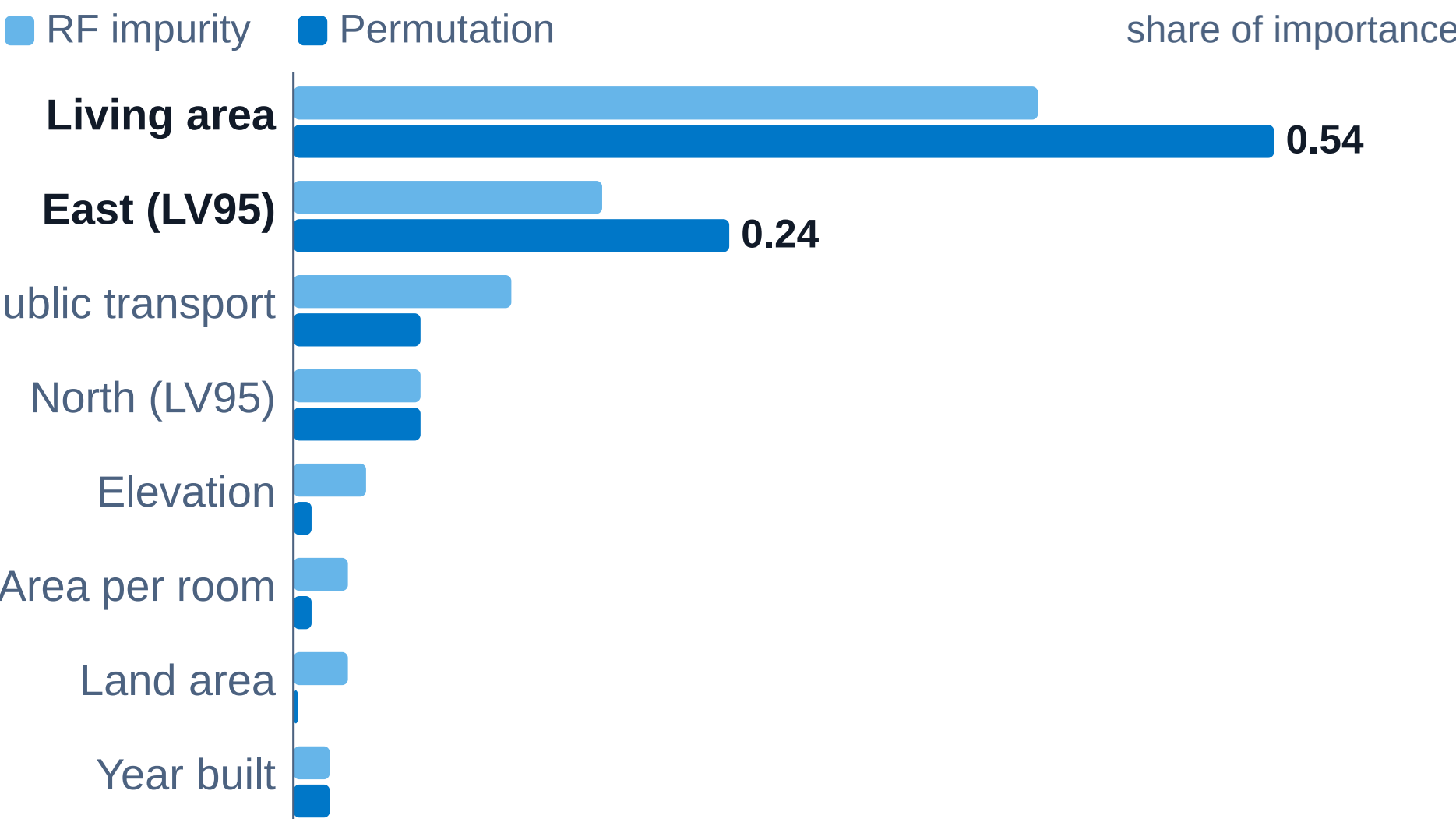
RMSE train vs. eval (CHF, engineered features)



LightGBM has the lowest eval RMSE (399), but memorises the training data (gap 249). Bootstrap intervals of the top tree models overlap, so the smaller gap decides.

04 WHAT DRIVES RENT?

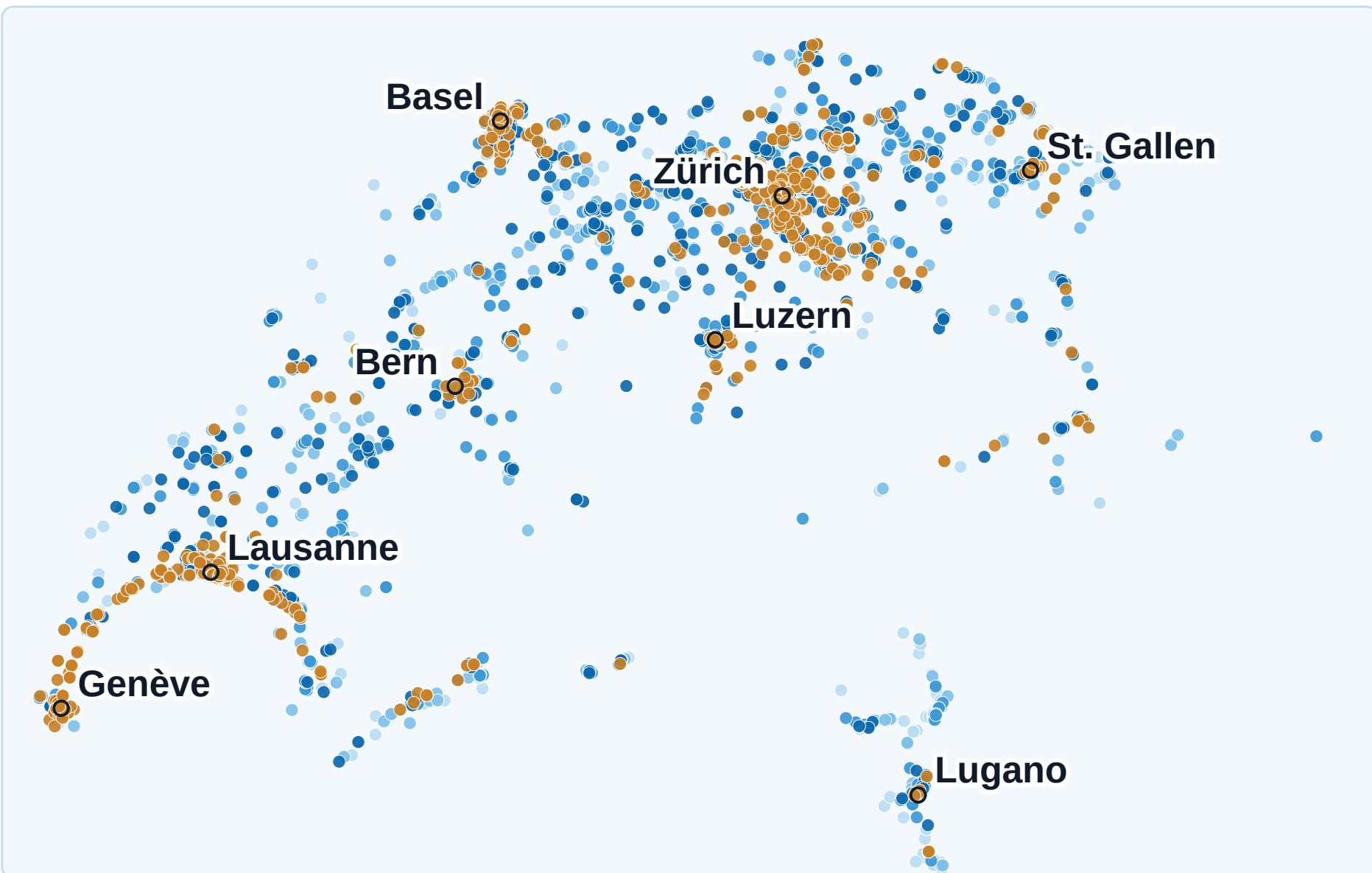
Living area is the strongest single predictor; location comes next.



Size and coordinates dominate every method. Public-transport access and elevation add local-market context.

05 WHERE IS RENT HIGH?

4'536 modelling apartments, coloured by monthly cold rent.



Zürich, Geneva, Basel and Lausanne form dense, high-rent clusters; urban markets are over-represented in the data.

06 TRY THE PROTOTYPE

Swiss address → official GWR dwelling → live rent estimate.

Live example: Baselstrasse 30, 6003 Luzern · 2nd floor, 50 m², 2 rooms



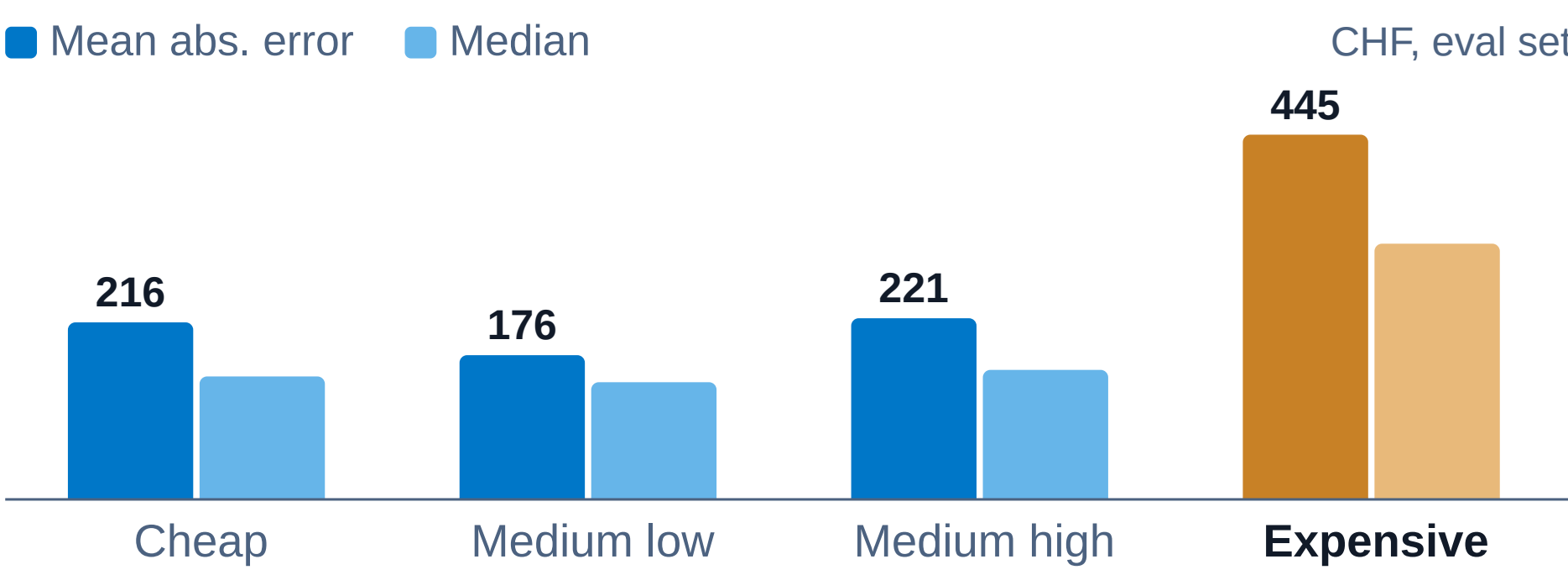
SCAN TO EXPLORE

Estimate a monthly cold rent for any Swiss address.

Demo model: LightGBM + KNN, eval RMSE 393

07 WHERE DOES IT STRUGGLE?

The most expensive quartile has about twice the error.



Missing signals such as view, floor, renovation state and finish matter most for premium flats, mainly around Zürich and Lake Geneva.

08 LIMITATIONS

- Snapshot dataset**
one scrape on 13 April 2026, no market trend
- Coverage bias**
urban listings over-represented, rural areas sparse
- Missing qualitative signals**
view, floor, renovation, balcony, furnishing
- High-price uncertainty**
largest errors for rare luxury apartments
- Overlapping model CIs**
the final choice is a robustness decision

09 NEXT STEPS

- Features from text + images**
extract floor, view and condition
- More listing platforms**
e.g. Homegate, ImmoScout24
- Repeated snapshots**
monthly scrapes instead of one
- Time-aware validation**
retrain and test on newer data
- Production deployment**
model versioning and drift monitoring

ACCURACY IS NOT EVERYTHING.

The final model was chosen because it balances accuracy, stability and explainability.